

Problem1

3.49. Let $x[n]$ be a periodic sequence with period N and Fourier series representation

$$x[n] = \sum_{k=\langle N \rangle} a_k e^{jk(2\pi/N)n}. \quad (\text{P3.49-1})$$

(a) Suppose that N is even and that $x[n]$ in eq. (P3.49-1) satisfies

$$x[n] = -x\left[n + \frac{N}{2}\right] \text{ for all } n.$$

Show that $a_k = 0$ for all even integers k .

(b) Suppose that N is divisible by 4. Show that if

$$x[n] = -x\left[n + \frac{N}{4}\right] \text{ for all } n,$$

then $a_k = 0$ for every value of k that is a multiple of 4.

(c) More generally, suppose that N is divisible by an integer M . Show that if

$$\sum_{r=0}^{(N/M)-1} x\left[n + r\frac{N}{M}\right] = 0 \text{ for all } n,$$

then $a_k = 0$ for every value of k that is a multiple of M .

Correction:

For part (c), the condition should be:

$$\sum_{r=0}^{M-1} x\left[n + r\frac{N}{M}\right] = 0, \quad \forall n.$$

Problem 1. $x[n] \rightarrow a_k$, $x[n-n_0] \rightarrow a_k e^{-jk\frac{2\pi}{N}n_0}$

$$(a) x[n] = -x\left[n - \frac{N}{2}\right] \Rightarrow a_k = -a_k e^{-jk\frac{2\pi}{N} \cdot \frac{N}{2}} = -a_k e^{-jk\pi} = -a_k (-1)^k$$

$$k \text{ 为偶数} \Rightarrow a_k = -a_k \Rightarrow a_k = 0.$$

两边取DTFS.

$$(b) x[n] = -x\left[n + \frac{N}{4}\right] \Rightarrow a_k = -a_k e^{jk\frac{2\pi}{N} \cdot \frac{N}{4}} = -a_k e^{jk\frac{\pi}{2}}$$

$$k \text{ 为4的倍数} \Rightarrow k=4r, a_k = -a_k e^{jk\frac{\pi}{2} \cdot 4r} = -a_k \Rightarrow a_k = 0.$$

$$(c) \sum_{r=0}^{M-1} x\left[n + r\frac{N}{M}\right] = 0$$

$$\Rightarrow \sum_{r=0}^{M-1} a_k e^{jk\frac{2\pi}{N} \cdot r\frac{N}{M}} = 0 \Rightarrow a_k \sum_{r=0}^{M-1} e^{jk\frac{2\pi}{M} r} = 0$$

$$k \text{ 是 } m \text{ 的倍数时, 令 } k=nm, a_k \sum_{r=0}^{M-1} e^{j2\pi \cdot nr} = 0 \Rightarrow a_k \sum_{r=0}^{M-1} 1 = 0 \Rightarrow a_k \cdot M = 0$$

$$\Rightarrow a_k = 0$$

Problem2

3.58. Let $x[n]$ and $y[n]$ be periodic signals with common period N , and let

$$z[n] = \sum_{r=-\infty}^{\infty} x[r]y[n-r]$$

be their periodic convolution.

(a) Show that $z[n]$ is also periodic with period N .

(b) Verify that if a_k , b_k , and c_k are the Fourier coefficients of $x[n]$, $y[n]$, and $z[n]$, respectively, then

$$c_k = Na_k b_k.$$

(c) Let

$$x[n] = \sin\left(\frac{3\pi n}{4}\right)$$

and

$$y[n] = \begin{cases} 1, & 0 \leq n \leq 3 \\ 0, & 4 \leq n \leq 7 \end{cases}$$

be two signals that are periodic with period 8. Find the Fourier series representation for the periodic convolution of these signals.

(d) Repeat part (c) for the following two periodic signals that also have period 8:

$$x[n] = \begin{cases} \sin\left(\frac{3\pi n}{4}\right), & 0 \leq n \leq 3 \\ 0, & 4 \leq n \leq 7 \end{cases}$$

$$y[n] = \left(\frac{1}{2}\right)^n, 0 \leq n \leq 7.$$

Problem2 证明 $z[n] = \sum_{r=-\infty}^{\infty} x[r]y[n-r]$

(a) 证: $z[n+N] = \sum_{r=-\infty}^{\infty} x[r]y[n+N-r] = \sum_{r=-\infty}^{\infty} x[r]y[n-r] = z[n]$
 $\Rightarrow z[n]$ 周期为 N .

(b) DFTs, $z[n] \rightarrow c_k$, $x[n] \rightarrow a_k$, $y[n] \rightarrow b_k$, 证 $c_k = Na_k b_k$

$$\begin{aligned} c_k &= \frac{1}{N} \sum_{n=-\infty}^{\infty} z[n] e^{-j\frac{2\pi}{N}kn} \\ &= \frac{1}{N} \sum_{n=-\infty}^{\infty} \left(\sum_{r=-\infty}^{\infty} x[r]y[n-r] \right) e^{-j\frac{2\pi}{N}kn} = \sum_{r=-\infty}^{\infty} x[r] \underbrace{\left(\frac{1}{N} \sum_{n=-\infty}^{\infty} y[n-r] e^{-j\frac{2\pi}{N}kn} \right)}_{b_k e^{-j\frac{2\pi}{N}kr}} \\ &= \sum_{r=-\infty}^{\infty} x[r] \cdot b_k e^{-j\frac{2\pi}{N}kr} \\ &= N \cdot b_k \cdot \sum_{r=-\infty}^{\infty} x[r] e^{-j\frac{2\pi}{N}kr} = Na_k b_k \end{aligned}$$

(c) 证 $x[n] = \sin\left(\frac{3\pi n}{4}\right)$, $y[n] = \begin{cases} 1, & 0 \leq n \leq 3 \\ 0, & 4 \leq n \leq 7 \end{cases}$ 求 c_k .

$$\text{对 } a_k: x[n] = \frac{1}{2j} (e^{j\frac{3\pi}{4}n} - e^{-j\frac{3\pi}{4}n}), N=8$$

$$\frac{1}{N} \sum_{n=-\infty}^{\infty} e^{j\frac{3\pi}{4}n} e^{-j\frac{2\pi}{8}kn} \Rightarrow k=3 \text{ 时, } \frac{1}{N} \sum_{n=-\infty}^{\infty} e^{j\frac{3\pi}{4}n} e^{-j\frac{2\pi}{8}kn} = 1.$$

$$\Rightarrow \begin{cases} a_3 = \frac{1}{2j} \\ a_5 = -\frac{1}{2j} \end{cases} \text{ 对 } [0, 7] \text{ 内其他 } k: a_k = 0.$$

$$\text{对 } b_k: b_k = \frac{1}{8} \sum_{n=0}^7 y[n] e^{-j\frac{2\pi}{8}kn} = \frac{1}{8} \sum_{n=0}^3 e^{-j\frac{2\pi}{8}kn} = \frac{1}{8} \frac{1-e^{-j\frac{2\pi}{8}k4}}{1-e^{-j\frac{2\pi}{8}k}}$$

$$\Rightarrow \begin{cases} b_3 = \frac{1}{8} \frac{1-e^{-j\pi}}{1-e^{-j\frac{\pi}{2}}} = \frac{1}{4(1-e^{-j\frac{\pi}{2}})} \\ b_5 = \frac{1}{8} \frac{1-e^{-j\frac{5\pi}{2}}}{1-e^{-j\frac{5\pi}{4}}} \end{cases}$$

$$\text{对 } c_k: \begin{cases} c_3 = 8 \cdot \frac{1}{2j} \cdot \frac{1}{4(1-e^{-j\frac{\pi}{2}})} = \frac{1}{j(1-e^{-j\frac{\pi}{2}})} \\ c_5 = \frac{-1}{j(1-e^{-j\frac{5\pi}{4}})} \end{cases} \text{ 对 } [0, 7] \text{ 内其他 } k: c_k = 0.$$

$$(d) \text{ 已知 } x[n] = \begin{cases} \sin(\frac{3\pi n}{4}), & 0 \leq n \leq 3 \\ 0, & 4 \leq n \leq 7 \end{cases} \quad y[n] = (\frac{1}{2})^n, \quad 0 \leq n \leq 7. \quad \text{求 } c_k.$$

$$\text{对 } a_k: a_k = \frac{1}{8} \sum_{n=0}^3 \sin(\frac{3\pi n}{4}) e^{-j\frac{k\pi}{4}n} = \frac{1}{16j} \sum_{n=0}^3 (e^{j\frac{3\pi}{4}n} - e^{-j\frac{3\pi}{4}n}) e^{-j\frac{k\pi}{4}n}$$

$$= \frac{1}{16j} \left[\sum_{n=0}^3 e^{j(3-k)\frac{\pi}{4}n} - \sum_{n=0}^3 e^{-j(3+k)\frac{\pi}{4}n} \right]$$

$$= \frac{1}{16j} \left[\frac{1 - e^{j(3-k)\pi}}{1 - e^{j(3-k)\frac{\pi}{4}}} - \frac{1 - e^{-j(3+k)\pi}}{1 - e^{-j(3+k)\frac{\pi}{4}}} \right], \quad k \neq 3, 5, \quad \begin{cases} a_3 = -j\frac{1}{4} \\ a_5 = j\frac{1}{4} \end{cases}$$

$$\text{对 } b_k: b_k = \frac{1}{8} \sum_{n=0}^7 (\frac{1}{2})^n e^{-j\frac{k\pi}{4}n} = \frac{1}{8} \sum_{n=0}^7 (\frac{1}{2} e^{-j\frac{k\pi}{4}})^n = \frac{1}{8} \frac{1 - \frac{1}{256} e^{-j\frac{k\pi}{4}}}{1 - \frac{1}{2} e^{-j\frac{k\pi}{4}}}$$

$$= \frac{255}{2048 (1 - \frac{1}{2} e^{-j\frac{k\pi}{4}})}$$

$$\text{对 } c_k: c_k = 8 a_k b_k = \frac{255}{4096 (1 - \frac{1}{2} e^{-j\frac{k\pi}{4}})} \left[\frac{1 - e^{j(3-k)\pi}}{1 - e^{j(3-k)\frac{\pi}{4}}} - \frac{1 - e^{-j(3+k)\pi}}{1 - e^{-j(3+k)\frac{\pi}{4}}} \right], \quad k \neq 3, 5.$$

$$c_3 = -\frac{255}{1024 (1 - \frac{1}{2} e^{-j\frac{3\pi}{4}})}$$

$$c_5 = \frac{255}{1024 (1 - \frac{1}{2} e^{-j\frac{5\pi}{4}})}$$

Problem3

- 4.25. Let $X(j\omega)$ denote the Fourier transform of the signal $x(t)$ depicted in Figure P4.25.
- Find $\angle X(j\omega)$.
 - Find $X(j0)$.
 - Find $\int_{-\infty}^{\infty} X(j\omega) d\omega$.
 - Evaluate $\int_{-\infty}^{\infty} X(j\omega) \frac{2 \sin \omega}{\omega} e^{j2\omega} d\omega$.
 - Evaluate $\int_{-\infty}^{\infty} |X(j\omega)|^2 d\omega$.
 - Sketch the inverse Fourier transform of $\text{Re}\{X(j\omega)\}$.
- Note: You should perform all these calculations without explicitly evaluating $X(j\omega)$.

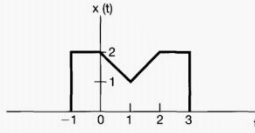
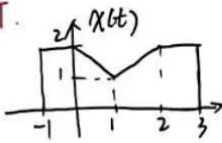


Figure P4.25

Problem3 已知 $x(t) \rightarrow X(j\omega)$

CTFT



(a) 求 $\angle X(j\omega)$

偶函数 $\rightarrow$ 纯实数

令 $g(t) = x(t+1)$, 则 $g(t)$ 关于 $t=0$ 对称, 为偶函数.

$\Rightarrow G(j\omega)$ 纯实数

时移

$$x(t) = g(t-1) \Rightarrow X(j\omega) = G(j\omega) e^{-j\omega} \Rightarrow \angle X(j\omega) = -\omega.$$

(b) 求 $X(j0)$

$$X(j\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt \Rightarrow X(j0) = \int_{-\infty}^{\infty} x(t) dt = 2 \times 4 - \frac{1}{2} \times 2 \times 1 = 7.$$

利用 $\omega=0$ 时
 $e^0 = 1$ 求
中面积.

(c) 求 $\int_{-\infty}^{\infty} X(j\omega) d\omega$

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) e^{j\omega t} d\omega \Rightarrow 2\pi \cdot x(0) = \int_{-\infty}^{\infty} X(j\omega) d\omega = 4\pi.$$

(d) 求 $\int_{-\infty}^{\infty} X(j\omega) \frac{2 \sin \omega}{\omega} e^{j2\omega} d\omega$

$$\Rightarrow X(j\omega) = \frac{2 \sin(\omega a)}{\omega}$$

$$\text{已知 } y(t) \Rightarrow Y(j\omega) = \frac{2 \sin \omega}{\omega}$$

$$\Rightarrow z(t) = y(t+2), z(j\omega) = \frac{2 \sin \omega}{\omega} \cdot e^{j2\omega}.$$

看成 γ 函数

$$\Rightarrow \text{求 } \int_{-\infty}^{\infty} X(j\omega) z(j\omega) d\omega$$

$$F^{-1}[X(j\omega) z(j\omega)] = x(t) * z(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) z(j\omega) e^{j\omega t} d\omega$$

$$t=0 \text{ 时, } x(0) * z(0) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) z(j\omega) d\omega$$

$$x(0) * z(0) = \int_{-\infty}^{\infty} x(\tau) z(-\tau) d\tau = \int_{-1}^3 x(\tau) d\tau = 3.5$$

$$\Rightarrow \int_{-\infty}^{\infty} X(j\omega) z(j\omega) d\omega = 2\pi \cdot 3.5 = 7\pi.$$

(e) 求 $\int_{-\infty}^{\infty} |X(j\omega)|^2 d\omega$

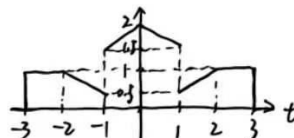
$$\int_{-\infty}^{\infty} |x(t)|^2 dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} |X(j\omega)|^2 d\omega$$

$$\int_{-\infty}^{\infty} |x(t)|^2 dt = \int_{-1}^0 4 dt + \int_0^1 (2-t)^2 dt + \int_1^2 t^2 dt + \int_2^3 4 dt = \frac{38}{3}$$

$$\Rightarrow \int_{-\infty}^{\infty} |X(j\omega)|^2 d\omega = 2\pi \cdot \frac{38}{3} = \frac{76}{3}\pi.$$

(f) 画 $F^{-1}[\text{Re}\{X(j\omega)\}]$ $x(t)$ 下偶部

$$F^{-1}[\text{Re}\{X(j\omega)\}] = x_e(t) = \frac{1}{2} (x(t) + x(-t))$$



Problem 4

4.37. Consider the signal $x(t)$ in Figure P4.37.

- Find the Fourier transform $X(j\omega)$ of $x(t)$.
- Sketch the signal

$$\tilde{x}(t) = x(t) * \sum_{k=-\infty}^{\infty} \delta(t - 4k).$$

- Find another signal $g(t)$ such that $\tilde{x}(t)$ is not the same as $x(t)$ and

$$\tilde{x}(t) = g(t) * \sum_{k=-\infty}^{\infty} \delta(t - 4k).$$

- Argue that, although $G(j\omega)$ is different from $X(j\omega)$, $G(j\frac{\pi k}{2}) = X(j\frac{\pi k}{2})$ for all integers k . You should not explicitly evaluate $G(j\omega)$ to answer this question.

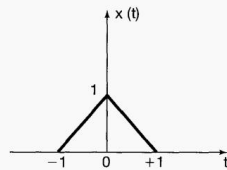
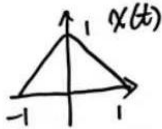


Figure P4.37

Problem 4



(a) 求 $X(j\omega)$

$$X(j\omega) = \int_{-1}^0 (1+t) e^{-j\omega t} dt + \int_0^1 (1-t) e^{-j\omega t} dt$$

$$\text{或: 令 } p(t) = \begin{cases} 1, & |t| < 0.5 \\ 0, & \text{其他} \end{cases} \quad \text{则 } x(t) = p(t) * p(t)$$

构造卷积, 用卷积性质.

$$P(j\omega) = \int_{-0.5}^{0.5} 1 \cdot e^{-j\omega t} dt = \frac{e^{-j0.5\omega} - e^{j0.5\omega}}{-j\omega} = \frac{2 \sin \frac{\omega}{2}}{\omega}$$

$$X(j\omega) = P(j\omega)^2 = \frac{4}{\omega^2} \sin^2 \left(\frac{\omega}{2} \right)$$

$$(b) \tilde{x}(t) = x(t) * \sum_{k=-\infty}^{\infty} \delta(t - 4k)$$

$$\Rightarrow \tilde{x}(t) = \sum_{k=-\infty}^{\infty} x(t - 4k)$$



$$(c) \text{找 } g(t): \text{使 } \tilde{x}(t) = g(t) * \sum_{k=-\infty}^{\infty} \delta(t - 4k)$$

$g(t)$ 可以是 $x(t)$ 平移 $T=4$ 的整数倍. eg. $g(t) = x(t-4)$.

(d) 证明, 尽管 $G(j\omega) \neq X(j\omega)$, 但 $G(j\frac{\pi k}{2}) = X(j\frac{\pi k}{2})$

$$\text{令 } y(t) = \sum_{k=-\infty}^{\infty} \delta(t - 4k)$$

$$\text{则 } x(t) * y(t) = g(t) * y(t)$$

$$\text{求 } Y(j\omega): Y(j\omega) = \frac{2\pi}{2} \sum_{k=-\infty}^{\infty} \delta(\omega - \frac{\pi k}{2})$$

$$\sum_{n=-\infty}^{\infty} \delta(t - nT) \xrightarrow{F} \sum_{n=-\infty}^{\infty} e^{-jn\omega T} = \frac{2\pi}{T} \sum_{k=-\infty}^{\infty} \delta(\omega - k\frac{2\pi}{T})$$

时域和周期冲激串, 频域还是周期冲激串.

$$X(j\omega) \cdot Y(j\omega) = G(j\omega) \cdot Y(j\omega)$$

$$f(x) \delta(x-a) = f(a) \delta(x-a)$$

$$\Rightarrow \frac{2\pi}{2} \sum_{k=-\infty}^{\infty} X(j\omega) \delta(\omega - \frac{\pi k}{2}) = \frac{2\pi}{2} \sum_{k=-\infty}^{\infty} X(j\frac{\pi k}{2}) \delta(\omega - \frac{\pi k}{2})$$

$$\Rightarrow \sum_{k=-\infty}^{\infty} X(j\frac{\pi k}{2}) \delta(\omega - \frac{\pi k}{2}) = \sum_{k=-\infty}^{\infty} Y(j\frac{\pi k}{2}) \delta(\omega - \frac{\pi k}{2})$$

$$\Rightarrow X(j\frac{\pi k}{2}) = Y(j\frac{\pi k}{2})$$

Problem5

5.19. Consider a causal and stable LTI system S whose input $x[n]$ and output $y[n]$ are related through the second-order difference equation

$$y[n] - \frac{1}{6}y[n-1] - \frac{1}{6}y[n-2] = x[n].$$

- (a) Determine the frequency response $H(e^{j\omega})$ for the system S .
 (b) Determine the impulse response $h[n]$ for the system S .

Problem5. $y[n] - \frac{1}{6}y[n-1] - \frac{1}{6}y[n-2] = x[n]$

(a) $Y(e^{j\omega}) - \frac{1}{6}e^{-j\omega}Y(e^{j\omega}) - \frac{1}{6}e^{-j2\omega}Y(e^{j\omega}) = X(e^{j\omega})$

$$\Rightarrow H(e^{j\omega}) = \frac{1}{1 - \frac{1}{6}e^{-j\omega} - \frac{1}{6}e^{-j2\omega}}$$

(b) $\hat{h}[n]$

$$H(e^{j\omega}) = \frac{0.6}{1 - \frac{1}{2}e^{-j\omega}} + \frac{0.4}{1 + \frac{1}{3}e^{-j\omega}}$$

$$\Rightarrow h[n] = \frac{3}{5}\left(\frac{1}{2}\right)^n u[n] + \frac{2}{5}\left(-\frac{1}{3}\right)^n u[n].$$

Problem6

5.34. Consider a system consisting of the cascade of two LTI systems with frequency responses

$$H_1(e^{j\omega}) = \frac{2 - e^{-j\omega}}{1 + \frac{1}{2}e^{-j\omega}}$$

and

$$H_2(e^{j\omega}) = \frac{1}{1 - \frac{1}{2}e^{-j\omega} + \frac{1}{4}e^{-j2\omega}}.$$

- (a) Find the difference equation describing the overall system.
 (b) Determine the impulse response of the overall system.

Problem6. $H_1(e^{j\omega}) = \frac{2 - e^{-j\omega}}{1 + \frac{1}{2}e^{-j\omega}}$, $H_2(e^{j\omega}) = \frac{1}{1 - \frac{1}{2}e^{-j\omega} + \frac{1}{4}e^{-j2\omega}}$

(a) $H(e^{j\omega}) = H_1(e^{j\omega}) \cdot H_2(e^{j\omega}) = \frac{2 - e^{-j\omega}}{1 + \frac{1}{8}e^{-j3\omega}}$

$$\Rightarrow Y(e^{j\omega}) \cdot (1 + \frac{1}{8}e^{-j3\omega}) = X(e^{j\omega}) \cdot (2 - e^{-j\omega})$$

$$\Rightarrow y[n] + \frac{1}{8}y[n-3] = 2x[n] - x[n-1]$$

(b) $\hat{h}[n]$

$$H(e^{j\omega}) = \frac{2}{1 + \frac{1}{8}e^{-j3\omega}} - \frac{e^{-j\omega}}{1 + \frac{1}{8}e^{-j3\omega}}$$

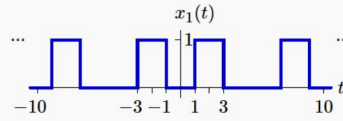
$$\hat{h}[n] \Rightarrow G(e^{j\omega}) = \frac{1}{1 + \frac{1}{8}e^{-j3\omega}} = \sum_{k=0}^{\infty} \left(-\frac{1}{8}\right)^k e^{-j3\omega k}$$

$$\Rightarrow g[n] = \sum_{k=0}^{\infty} \left(-\frac{1}{8}\right)^k \delta[n-3k]$$

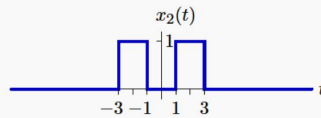
$$\Rightarrow h[n] = 2g[n] - g[n-1] = 2 \sum_{k=0}^{\infty} \left(-\frac{1}{8}\right)^k \delta[n-3k] - \sum_{k=0}^{\infty} \left(-\frac{1}{8}\right)^k \delta[n-3k-1]$$

Problem 7:

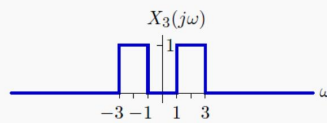
- a. Determine the Fourier series coefficients of the following signal, which is periodic in $T = 10$.



- b. Determine the Fourier transform of the following signal, which is zero outside the indicated range.



- c. Determine the time waveform that corresponds to the following Fourier transform, which is zero outside the indicated range.



What is the relation between the answer to this part and that of the previous part?

Problem 7. (a) $x_1(t)$ $\bar{T}=10$. 求 a_k .
 $\omega_0 = \frac{2\pi}{T} = \frac{2\pi}{10} = \frac{\pi}{5}$, $x_1(t) = \begin{cases} 1, & |t| \in (1, 3) \\ 0, & \text{其他} \end{cases}$
 $\frac{1}{2} p(t)$: $a_{pk} = \frac{\sin k \cdot \frac{\pi}{5} \cdot 1}{k\pi} = \frac{\sin(\frac{k\pi}{5})}{k\pi}$
 $x_1(t) = p(t+2) + p(t-2) \Rightarrow a_k = a_{pk}(e^{j2k\omega_0} + e^{-j2k\omega_0}) = \frac{2}{k\pi} \sin(\frac{k\pi}{5}) \cos(\frac{2k\pi}{5})$
 $a_0 = \frac{1}{10} \times (2+2) = \frac{2}{5}$, $\lim_{k \rightarrow 0} \frac{2}{k\pi} \sin(\frac{k\pi}{5}) \cos(\frac{2k\pi}{5}) = \frac{2}{5}$ 相同
 $\Rightarrow a_k = \frac{2}{k\pi} \sin(\frac{k\pi}{5}) \cos(\frac{2k\pi}{5})$

(b) $x_2(t)$ 非周期. 求 $x_2(j\omega)$
 $P(j\omega) = \frac{2\sin\omega}{\omega}$, $x_2(t) = p(t+2) + p(t-2)$
 $\Rightarrow x_2(j\omega) = P(j\omega)e^{j2\omega} + P(j\omega)e^{-j2\omega} = P(j\omega) \cdot 2\cos 2\omega$
 $= \frac{4}{\omega} \sin\omega \cos 2\omega$

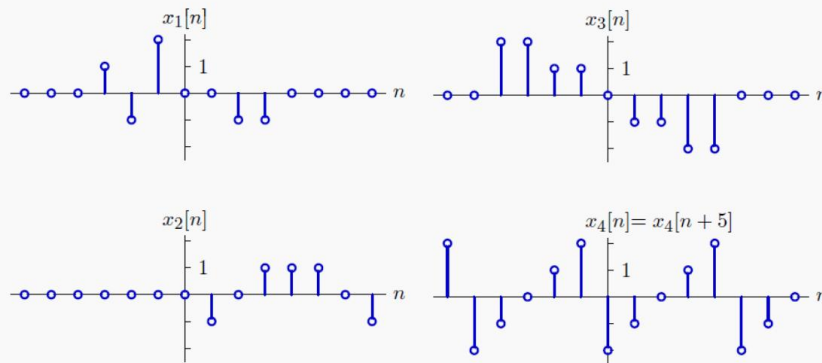
(c) $x_3(j\omega)$ ~~$x_3(t)$~~ $x_3(j\omega) = x_2(\omega)$
 $x_2(t) \xrightarrow{F} x_2(j\omega)$
 $x_2(jt) \xrightarrow{F} 2\pi x_2(-\omega)$
 $\frac{1}{2\pi} x_2(jt) \xrightarrow{F} x_2(-\omega) = x_2(\omega) = x_3(j\omega)$
 $\Rightarrow x_3(t) = \frac{2}{\pi t} \sin t \cos 2t$

Problem 8:

Which are True?

For each of the DT signals $x_1[n]$ through $x_4[n]$ (below), determine whether the conditions listed in the following table are satisfied, and answer **T** for true or **F** for false.

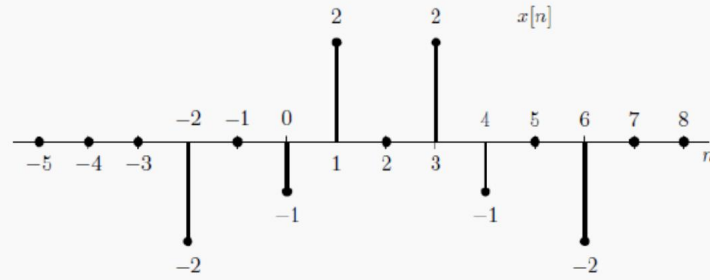
	$x_1[n]$	$x_2[n]$	$x_3[n]$	$x_4[n]$
$X(e^{j0}) = 0$				
$\int_{-\pi}^{\pi} X(e^{j\Omega}) d\Omega = 0$				
$X(e^{j\Omega})$ is purely imaginary				
$e^{jk\Omega} X(e^{j\Omega})$ is purely real for some integer k				



- Problem 8. (1) $X(e^{j0})=0 \Rightarrow \sum_{n=-\infty}^{\infty} x[n]=0 \Rightarrow x_1, x_3$ 满足
- (2) $\int_{-\pi}^{\pi} X(e^{j\Omega}) d\Omega = 0 \Rightarrow x[0]=0 \Rightarrow x_1, x_3$ 满足
- (3) $X(e^{j\Omega})$ 纯虚数 $\Rightarrow x[n]$ 奇函数 $\Rightarrow x_3$ 满足.
- (4) $e^{jk\Omega} X(e^{j\Omega})$ 对某整数 k , 结果是实数. $\Rightarrow$ 关于 $n=-k$ 对称. $\Rightarrow x_2$ 满足.

Problem 9:

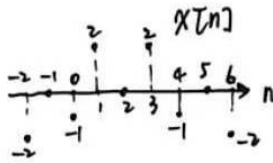
Let $X(e^{j\omega})$ denote the Fourier transform of the signal $x[n]$ depicted below.



- Find $X(1) = X(e^{j0})$.
- Find α such that $e^{j\alpha\omega} X(e^{j\omega})$ is real.
- Evaluate $\int_{-\pi}^{\pi} X(e^{j\omega}) d\omega$.
- Find $X(e^{j\pi})$.
- Determine and sketch the signal whose Fourier transform is $\Re\{X(e^{j\omega})\}$.
- Evaluate the following integrals:

$$\int_{-\pi}^{\pi} |X(e^{j\omega})|^2 d\omega$$

Problem 9.



$$(a) X(e^{j0}) = \sum_{n=-\infty}^{\infty} x[n] = -2$$

(b) 找到使 $e^{j\alpha\omega} X(e^{j\omega})$ 为实数

$$F^{-1}[e^{j\alpha\omega} X(e^{j\omega})] = x[n+\alpha] \Rightarrow \alpha=2.$$

$$(c) \int_{-\pi}^{\pi} X(e^{j\omega}) d\omega$$

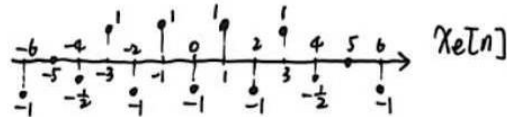
$$X(e^{j0}) = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega}) d\omega, \quad \text{所以} = 2\pi X(e^{j0}) = -2\pi.$$

$$(d) X(e^{j\pi})$$

$$X(e^{j\pi}) = \sum_{n=-\infty}^{\infty} x[n] \cdot e^{-j\pi n} = \sum_{n=-\infty}^{\infty} x[n] \cdot (-1)^n = x[-2] + x[0] - x[1] - x[3] + x[4] + x[6] = -10$$

$$(e) F^{-1}[\Re\{X(e^{j\omega})\}]$$

$$\Rightarrow x_e[n] = \frac{1}{2} (x[n] + x[-n])$$



$$(f) \int_{-\pi}^{\pi} |X(e^{j\omega})|^2 d\omega$$

$$\sum_{n=-\infty}^{\infty} |x[n]|^2 = \frac{1}{2\pi} \int_{-\pi}^{\pi} |X(e^{j\omega})|^2 d\omega$$

$$\Rightarrow \text{所以} = 2\pi (4+1+4+4+1+4) = 36\pi.$$